04.07.15
Saturday

Unit - II

Conformal mapping

2m
(1)

A mapping that preserves angles between any oriented curves both in magnitude and in direction is called as an conformal mapping.

(1) 2m Isogonal mapping:

The mapping is said to be isogonal if it preserves the magnitude of the angles but not necessarily the direction

Critical point:

A point in which $f'(z)=0$ is called a critical point of the transformation.

1) Find the critical points of the following transformation

6 times
2m
(1)

(1) $w^2=(z-\alpha)(z-\beta)$

(1)

$\boxed{w=f(z)=u+iv}$

$w^2=z^2-z\beta-z\alpha+\alpha\beta$

$w^2=z^2-z(\alpha+\beta)+\alpha\beta$

Diff w.r.to z on both sides.

$2w\frac{dw}{dz}=2z-(\alpha+\beta)+0$

$f'(z)=\frac{dw}{dz}=\frac{2z-(\alpha+\beta)}{2w}$

$f'(z)=0$

$\frac{2z-(\alpha+\beta)}{2w}=0$

$2z = \alpha + \beta$

$$\boxed{z = \frac{\alpha+\beta}{2}}$$

06.07.15
Monday

1) Transformation ① $w = c + z$

Rep
① 2m

where $w = u + iv$

$c = a + ib$

$z = x + iy$.

$$u + iv = a + ib + x + iy$$

$$(u + iv) = (a + x) + i(b + y)$$

$$u = a + x \qquad v = b + y$$

The transformation (x, y) in the z plane is transformed into $u = a + x$ and $v = b + y$ in ~~the~~ w plane

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\cos iy = \cosh y$

$\sin iy = i \sinh y$

Transformation 2

6m
①

$$w = \sin z$$

$$u + iv = \sin(x + iy)$$

$$u + iv = \sin x \cos iy + \cos x \sin iy.$$

$$u + iv = \sin x \cosh y + i \sinh y \cos x.$$

$$u = \sin x \cosh y \; — ① \qquad v = \cos x \sinh y \; — ②$$

Eliminate x from ① & ②, we get

$$\sin x = \frac{u}{\cosh y}$$

$$\cos x = \frac{v}{\sinh y}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$$

$\frac{u^2}{a^2}+\frac{v^2}{b^2}=1$ | $\left[\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\right]$

which represents an ellipse. | $\cosh^2 y - \sinh^2 y = 1$

From ① & ② eliminate y we get.

$\cosh y = \frac{u}{\sin x}$

$\sinh y = \frac{-v}{\cos x}$

$\cosh^2 y - \sinh^2 y = 1$

$\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1$

$\frac{u^2}{a^2} - \frac{v^2}{b^2} = 1$

which represents hyperbola.

$\cos(A+B) = \cos A\cos B - \sin A \sin B$

Transformation 3:

$\cos iy = \cosh y$

$\sin iy = i\sinh y$.

$w = \cos z$.

$u+iv = \cos(x+iy)$.

$u+iv = \cos x\cos iy - \sin x \sin iy$

$u+iv = \cos x\cosh y - i\sin x\sinh y$.

$u = \cos x\cosh y$ — ① $\quad v = -\sin x\sinh y$ — ②

eliminate x from ① & ②

$\cos x = \frac{u}{\cosh y}$

$-\sin x = \frac{-v}{\sinh y}$.

$\cos^2 x + \sin^2 x = 1$

$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$

$$\frac{u^2}{a^2}+\frac{v^2}{b^2}=1$$

which represents an ellipse.

From ① & ② eliminate y we get.

$$\cosh y=\frac{u}{\cos x}$$

$$\sinh y=-\frac{v}{\sin x}$$

$$\cosh^2 y-\sinh^2 y=1$$

$$\frac{u^2}{\cos^2 x}-\frac{v^2}{\sin^2 x}=1 \quad \left[\because \frac{u^2}{a^2}-\frac{v^2}{b^2}=1\right]$$

which represents an hyperbola

Transformation 4:

5 times

$$w=z+\frac{1}{z}$$

Replace $z=re^{i\theta}$

$$w=re^{i\theta}+\frac{1}{re^{i\theta}}$$

$$w=re^{i\theta}+\frac{1}{r}(e^{-i\theta})$$

$$w=r(\cos\theta+i\sin\theta)+\frac{1}{r}(\cos\theta-i\sin\theta)$$

$$u+iv=r\cos\theta+ir\sin\theta+\frac{1}{r}\cos\theta-\frac{i}{r}\sin\theta$$

$$u+iv=r\cos\theta+\frac{1}{r}\cos\theta+i\left(r\sin\theta-\frac{1}{r}\sin\theta\right)$$

$$u=r\cos\theta+\frac{1}{r}\cos\theta \qquad v=r\sin\theta-\frac{1}{r}\sin\theta$$

$$u=\cos\theta\left(r+\frac{1}{r}\right) \quad -① \qquad v=\sin\theta\left(r-\frac{1}{r}\right) \quad -②$$

Eliminate θ from ① & ②

$$\cos\theta=\frac{u}{(r+1/r)}$$

$$\sin\theta=\frac{v}{(r-1/r)}$$

$\cos^2\theta + \sin^2\theta = 1$

$$\frac{u^2}{(x+1/x)^2} + \frac{v^2}{(x-1/x)^2} = 1$$

$\frac{u^2}{a^2} + \frac{v^2}{b^2} = 1$ where $a = x+1/x$, $b = x - 1/x$

which represents an ellipse

Eliminate x from ③ & ④

$$\left(x+\frac{1}{x}\right) = \frac{u}{\cos\theta} \quad \text{—③}$$

$$\left(x-\frac{1}{x}\right) = \frac{v}{\sin\theta} \quad \text{—④}$$

Square & subtract.

$$\left(x+1/x\right)^2 - \left(x-\frac{1}{x}\right)^2 = \frac{u^2}{\cos^2\theta} - \frac{v^2}{\sin^2\theta}$$

$$\left(x+1/x+x-1/x\right)\left(\cancel{x}+\frac{1}{x}-\cancel{x}+\frac{1}{x}\right) = \frac{u^2}{\cos^2\theta} - \frac{v^2}{\sin^2\theta}$$

$$4 = \frac{u^2}{\cos^2\theta} - \frac{v^2}{\sin^2\theta} \Rightarrow \frac{u^2}{4\cos^2\theta} - \frac{v^2}{4\sin^2\theta} = 1$$

$$\frac{u^2}{a^2} - \frac{v^2}{b^2} = 1$$

which represents an hyperbola.

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07.07.15
Tuesday

Transformation 5:

$$w = z + \frac{k^2}{z}$$

Replace z by $re^{i\theta}$

$$w = re^{i\theta} + \frac{k^2}{re^{i\theta}}$$

$$u+iv = re^{i\theta} + \frac{k^2 e^{-i\theta}}{r}$$

$$u+iv = r(\cos\theta + i\sin\theta) + \frac{k^2}{r}(\cos\theta - i\sin\theta)$$

$$u+iv = r\cos\theta + ir\sin\theta + \frac{k^2}{r}\cos\theta - i\frac{k^2}{r}\sin\theta.$$

$$u+iv = \left(r\cos\theta + \frac{k^2}{r}\cos\theta\right) + i\left(r\sin\theta - \frac{k^2}{r}\sin\theta\right)$$

$$u = r\cos\theta + \frac{k^2}{r}\cos\theta \quad \text{--- (1)}$$

$$v = r\sin\theta - \frac{k^2}{r}\sin\theta \quad \text{--- (2)}$$

Eliminate θ from (1) & (2).

$$\cos\theta = \frac{u}{\left(r + \frac{k^2}{r}\right)} \quad \text{--- (3)}$$

$$\sin\theta = \frac{v}{\left(r - \frac{k^2}{r}\right)} \quad \text{--- (4)}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\frac{u^2}{\left(r+\frac{k^2}{r}\right)^2} + \frac{v^2}{\left(r-\frac{k^2}{r}\right)^2} = 1$$

$$\frac{u^2}{a^2} + \frac{v^2}{b^2} = 1$$

which represents an ellipse.

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Eliminate r from (3) & (4)

$$\left(r+\frac{k^2}{r}\right)=\frac{u}{\cos\theta} \quad -(5)$$

$$\left(r-\frac{k^2}{r}\right)=\frac{v}{\sin\theta}. \quad -(6)$$

squaring & subtract (5) and (6)

$$\left(r+\frac{k^2}{r}\right)^2-\left(r-\frac{k^2}{r}\right)^2=\frac{u^2}{\cos^2\theta}-\frac{v^2}{\sin^2\theta}$$

$$\left(r+\cancel{\frac{k^2}{r}}+r-\cancel{\frac{k^2}{r}}\right)\left(\cancel{r}+\frac{k^2}{r}-\cancel{r}+\frac{k^2}{r}\right)=\frac{u^2}{\cos^2\theta}-\frac{v^2}{\sin^2\theta}$$

$$2\cancel{r}\times\frac{2k^2}{\cancel{r}}=\frac{u^2}{\cos^2\theta}-\frac{v^2}{\sin^2\theta}.$$

$$\frac{u^2}{4k^2\cos^2\theta}-\frac{v^2}{4k^2\sin^2\theta}=1$$

which represents an hyperbola.

1) Discuss the transformation $w=\frac{1}{z}$

11 m 10 times (2)

(or)

$$z=1/w$$

put $z=x+iy$

and $w=u+iv$.

$$w=\frac{1}{z}$$

$$u+iv=\frac{1}{x+iy}$$

$$u+iv=\frac{1}{x+iy}\times\frac{x-iy}{x-iy}$$

$$u+iv = \frac{x-iy}{x^2+y^2}$$

$$u = \frac{x}{x^2+y^2} \quad (1), \quad v = \frac{-y}{x^2+y^2} \quad (2)$$

Let $z = \frac{1}{w}$

$$x+iy = \frac{1}{u+iv}$$

$$x+iy = \frac{1}{u+iv} \times \frac{u-iv}{u-iv}$$

$$x+iy = \frac{u-iv}{u^2+v^2}$$

$$x = \frac{u}{u^2+v^2} \quad (3) \qquad y = \frac{-v}{u^2+v^2} \quad (4)$$

The general equation of a circle is $x^2+y^2+2gx+2fy+c=0$ — (1)

$$\left(\frac{u}{u^2+v^2}\right)^2 + \left(\frac{-v}{u^2+v^2}\right)^2 + 2g\left(\frac{u}{u^2+v^2}\right) + 2f\left(\frac{-v}{u^2+v^2}\right) + c = 0$$

$$\frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} + 2g\left(\frac{u}{u^2+v^2}\right) + 2f\left(\frac{-v}{u^2+v^2}\right) + c = 0$$

$$\frac{u^2+v^2 + 2gu(u^2+v^2) - 2fv(u^2+v^2) + c(u^2+v^2)^2}{(u^2+v^2)^2} = 0$$

$$(u^2+v^2) + 2gu(u^2+v^2) - 2fv(u^2+v^2) + c(u^2+v^2)^2 = 0$$

$$(u^2+v^2)\left[1 + 2gu - 2fv + c(u^2+v^2)^2\right] = 0$$

straight line equation

$$\boxed{\begin{array}{c} ax+by+c=0 \\ au+bv+c=0 \end{array}}$$

$$2gu - 2fv + 1 = 0$$

when $c=0$ in (1)

$$x^2+y^2+2gx+2fy = 0$$

replace $c=0$ in (2)

$$(u^2+v^2) + 2gu(u^2+v^2) - 2fv(u^2+v^2) = 0$$

$$1 + 2gu - 2fv = 0$$

$\therefore$ It represents a straight line.

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3) Find the image of circle $|z-1|=1$ under the mapping $w=\frac{1}{z}$

Soln $|z-1|=1$

$|(x+iy)-1|=1$

$|(x-1)+iy|=1$
(z)

$\sqrt{[(x-1)+iy][(x-1)-iy]}=1$
$(z)\quad(\bar{z})$

$\sqrt{(x-1)^2-(iy)^2}=1$

$\sqrt{(x-1)^2+y^2}=1$

Squaring on both sides

$(x-1)^2+y^2=(1)^2$

$(x-1)^2+y^2=1$

$x^2+1-2x+y^2=1$

$x^2+y^2-2x+\cancel{1}-\cancel{1}=0$

$x^2+y^2-2x=0$ —— (1)

$|z|=1$

$\sqrt{z\bar{z}}=1$

$\sqrt{(x+iy)(x-iy)}=1$

$\sqrt{x^2-(iy)^2}=1$

$\sqrt{x^2+y^2}=1$

Equations of circle

$(x-a)^2+(y-b)^2=r^2$

$x^2+y^2+2gx+2fy+c=0$

wKT

$x=\frac{u}{u^2+v^2}$, $y=\frac{-v}{u^2+v^2}$

From (1)

$\frac{u^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}-\frac{2u}{u^2+v^2}=0$

$\frac{u^2+v^2-2u(u^2+v^2)}{(u^2+v^2)^2}=0$

$u^2+v^2-2u(u^2+v^2)=0$

$(u^2+v^2)(1-2u)=0$

$1-2u=0$

$2u=1$

$\boxed{u=1/2}$

which represents a straight line.

3) Find the Image of the circle $|z+1|=1$ under the

mapping $w=\frac{1}{z}$

11 m
~~mobile~~
transfor

$|z+1|=1$

$|x+iy+1|=1$

$|(x+1)+iy|=1$

$\sqrt{[(x+1)+iy][(x+1)-iy]}=1$

$\sqrt{(x+1)^2+y^2}=1$

$(x^2+1)^2+y^2=1^2$

$x^2+\cancel{1}+2x+y^2\cancel{-1}=0$

$x^2+y^2+2x=0 \longleftarrow (1)$

WKT

$x=\frac{u}{u^2+v^2}, \; y=\frac{-v}{u^2+v^2}$

Sub x, y in (1), we get.

$\frac{u^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}+\frac{2u}{u^2+v^2}=0$

$\frac{u^2+v^2+2u(u^2+v^2)}{(u^2+v^2)^2}=0$

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$$u^2+v^2+2u(u^2+v^2)=0$$

$$(u^2+v^2)(1+2u)=0$$

$$1+2u=0$$

$$2u=-1$$

$$\boxed{u=-\tfrac{1}{2}}$$

which represents a straight line

4) Find the image of the circle $|z-2i|=2$ under the mapping $w=\frac{1}{z}$

$$|x+iy-2i|=2$$

$$|x+i(y-2)|=2$$

$$\sqrt{[x+i(y-2)][x-i(y-2)]}=2$$

$$\sqrt{x^2+(y-2)^2}=2$$

$$x^2+(y-2)^2=4.$$

$$x^2+y^2+\cancel{4}-4y-\cancel{4}=0$$

$$x^2+y^2-4y=0 \longleftarrow (1)$$

WKT

$$x=\frac{u}{u^2+v^2}, \quad y=\frac{-v}{u^2+v^2}$$

$$\frac{u^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}+\frac{4v}{u^2+v^2}=0$$

$$\frac{\cancel{(u^2+v^2)}}{(u^2+v^2)^{\cancel{2}}}+\frac{4v}{(u^2+v^2)}=0$$

$$\frac{1}{(u^2+v^2)}+\frac{4v}{(u^2+v^2)}=0$$

$$1+4v=0$$

$$4v=-1$$

$$\boxed{v=-1/4}$$

which represents a straight line

5) Find the image of the circle $|z+2i|=2$ under the mapping $w=1/z$

$$|x+iy+2i|=2$$

$$|x+i(y+2)|=2$$

$$\sqrt{[x+i(y+2)][x-i(y+2)]}=2$$

$$\sqrt{x^2+(y+2)^2}=2$$

$$x^2+(y+2)^2=4$$

$$x^2+y^2+4y+\cancel{4}-\cancel{4}=0$$

$$x^2+y^2+4y=0$$

WKT

$$x=\frac{u}{u^2+v^2},\ y=\frac{-v}{u^2+v^2}$$

$$\frac{u^2}{(u^2+v^2)^2}+\frac{v^2}{(u^2+v^2)^2}-\frac{4v}{(u^2+v^2)}=0$$

$$\frac{\cancel{(u^2+v^2)}}{(u+v^2)^{\cancel{2}}}-\frac{4v}{(u^2+v^2)}=0$$

$$\frac{1}{(u^2+v^3)} - \frac{4v}{(u^2+v^2)} = 0$$

$$1 = 4v$$

$$\boxed{v = 1/4}$$

which represents a straight line

6) Find the image of the line $y-x+1=0$ under the mapping $w = \frac{1}{z}$.

4 Find

WKT

$$x = \frac{u}{u^2+v^2}, \quad y = \frac{-v}{u^2+v^2}$$

$$\frac{-v}{u^2+v^2} - \frac{u}{u^2+v^2} + 1 = 0$$

$$\frac{-v-u+u^2+v^2}{u^2+v^2} = 0$$

$$u^2+v^2-(v+u) = 0$$

$$(u^2-u)+(v^2-v) = 0$$

$$(u-1/2)^2+(-1/4)+(v-1/2)^2+(-1/4) = 0$$

$$(u-1/2)^2+(v-1/2)^2 - \frac{1}{4} - \frac{1}{4} = 0$$

$$(u-1/2)^2+(v-1/2)^2 = +2/4$$

$$(u-1/2)^2+(v-1/2)^2 = 1/2$$

which represents a circle

Formula to make perfect square

$(x+A)^2 + B^2$

where $A = \frac{\text{coefficient of } x}{2}$

$B^2 = \text{constant} - (A^2)$

$(u-A)^2 + B^2$

where $A = -1/2$

$B^2 = 0 - (-1/2)^2$

$= 0 - 1/4 = -1/4$

7) Show that the transformation $w = \frac{iz+1}{z+i}$ maps the circle $|z|=1$ in the z plane into the real axis of the w plane

08.07.15 Wednesday 2 Marks 6m

Gn

$|z| = 1$

$w(z+i) = iz+1$

$wz + iw = iz+1$

$wz - iz = 1 - iw$

$z(w-i) = 1 - iw$

$z = \frac{1-iw}{w-i}$

$|z| = 1$

$\left| \frac{1-iw}{w-i} \right| = 1$

$\frac{|1-iw|}{|w-i|} = 1$

$|1-iw| = |w-i|$

$|1-i(u+iv)| = |u+iv-i|$ $[\because w = u+iv]$

$|1-iu+v| = |u+i(v-1)|$

$|(v+1)-iu| = |u+i(v-1)|$

$\sqrt{[(v+1)-iu][(v+1)+iu]} = \sqrt{[u+i(v-1)][u-i(v-1)]}$

$\sqrt{(v+1)^2+u^2} = \sqrt{u^2+(v-1)^2}$

$(v+1)^2 + \cancel{u^2} = \cancel{u^2} + (v-1)^2$

$v^2+2v+1 = v^2-2v+1$

$\cancel{v^2}+2v+\cancel{1}-\cancel{v^2}+2v-\cancel{1} = 0$

$4v = 0$

7) Show that the transformation $w = \frac{iz+1}{z+i}$ maps the circle $|z|=1$ in the z plane into the real axis of the w plane

08-07-15
Wednesday
2 times
6m Gn

$|z| = 1$

$w(z+i) = iz+1$

$wz + iw = iz + 1$

$wz - i\cancel{z} = 1 - iw$

$z(w-i) = 1 - iw$

$z = \frac{1-iw}{w-i}$

$|z| = 1$

$\left| \frac{1-iw}{w-i} \right| = 1$

$\frac{|1-iw|}{|w-i|} = 1$

$|1-iw| = |w-i|$

$|1-i(u+iv)| = |u+iv-i|$ $[\because w = u+iv]$

$|1-iu+v| = |u+i(v-1)|$

$|(v+1) - iu| = |u + i(v-1)|$

$\sqrt{[(v+1)-iu][(v+1)+iu]} = \sqrt{[u+i(v-1)][u-i(v-1)]}$

$\sqrt{(v+1)^2+u^2} = \sqrt{u^2+(v-1)^2}$

$(v+1)^2 + \cancel{u^2} = \cancel{u^2} + (v-1)^2$

$v^2 + 2v + 1 = v^2 - 2v + 1$

$\cancel{v^2} + 2v + \cancel{1} - \cancel{v^2} + 2v - \cancel{1} = 0$

$4v = 0$

$v = 0$

which represents a ~~straight line~~ the real axis of a w plane

5 8) Find the image of the infinite strips

① (i) $1/4 < y < 1/2$ (ii) $0 < y < 1/2$ under the transformation

② $w = 1/2$

wkT,

$$x = \frac{u}{u^2+v^2}, \quad y = \frac{-v}{u^2+v^2}$$

(i) $1/4 < y < 1/2$

$$\frac{1}{4} = y = \frac{1}{2}$$

Let $y = 1/4$ and $y = 1/2$

$$1/4 = \frac{-v}{u^2+v^2} \quad \left[y = \frac{-v}{u^2+v^2}\right]$$

$$u^2+v^2 = -4v$$

$$u^2+v^2+4v=0$$

$$u^2+v^2+4v+4-4 = 0$$

$$u^2+(v+2)^2-4 = 0$$

$$u^2+(v+2)^2 = 4$$

It represents a circle of radius 2 and with centre $(0,-2)$

$$\frac{1}{2} = \frac{-v}{u^2+v^2}$$

$$u^2+v^2 = -2v$$

$$u^2+v^2+2v=0$$

$$u^2+v^2+2v+1-1=0$$

$$u^2+(v+1)^2-1=0$$

$u^2+(v+1)^2=1$

It represents a circle of radius 1 and center $(0,-1)$

$u^2+(v+2)^2=4 < y < u^2+(v+1)^2=1.$

(ii) $0<y<\frac{1}{2}$.

$0=y=\frac{1}{2}$.

$0=\frac{-v}{u^2+v^2}$

(or) $u^2+v^2=-v$

$v=0$

It represents the real axis of a w plane.

$\frac{1}{2}=\frac{-v}{u^2+v^2}$

$u^2+v^2=-2v$

$u^2+v^2+2v=0$

$u^2+v^2+2v+1-1=0.$

$u^2+(v+1)^2=1$

It represents a circle of radius 1 and center $(0,-1)$

$0<y<u^2+(v+1)^2=1.$

9) The transformation $w=\sinh z$

Replace $w=u+iv$

$z=x+iy$.

$u+iv=\sinh(x+iy)$

$\left[\begin{array}{l}\sin iz = i\sinh z\\ \sinh z=\frac{\sin iz}{i}\end{array}\right]$

$u+iv=\frac{1}{i}\sin i(x+iy)$

$u+iv=\frac{1}{i}\sin(ix-y)$

$u+iv=\frac{1}{i}[\sin ix\cos y-\cos ix\sin y]$

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$u+iv = \frac{1}{i}\left[i \sinh x \cos y - \cosh x \sin y\right]$; $\sin iz = i\sinh z$, $\cos iz = \cosh z$

$u+iv = \frac{1}{i}\left[i \sinh x \cos y - \cosh x \sin y\right]$

$= \frac{i}{i} \sinh x \cos y - \frac{1}{i} \cosh x \sin y.$

$= \sinh x \cos y + \frac{i \times i}{i \times -i} \cosh x \sin y.$

$u+iv = \sinh x \cos y + \frac{i}{1} \cosh x \sin y.$

$u+iv = \sinh x \cos y + i \cosh x \sin y.$

$u = \sinh x \cos y$ ——(1) $v = \cosh x \sin y$ ——(2)

From (1) & (2) eliminate x.

$\cosh^2 x - \sinh^2 x = 1 = \sin^2 x + \cos^2 x$

$\sinh x = \frac{u}{\cos y}$

$\cosh x = \frac{v}{\sin y}.$

$\cosh^2 x - \sinh^2 x = 1$

$\frac{v^2}{\sin^2 y} - \frac{u^2}{\cos^2 y} = 1$

which represents a hyperbola.

From (1) & (2) eliminate y.

$\cos y = \frac{u}{\sinh x}$

$\sin y = \frac{v}{\cosh x}.$

$u+iv = \frac{1}{i}\left[i \sinh x \cos y - \cosh x \sin y\right]$

$u+iv = \frac{1}{i}\left[i \sinh x \cos y - \cosh x \sin y\right]$

$= \frac{i}{i} \sinh x \cos y - \frac{1}{i} \cosh x \sin y.$

$= \sinh x \cos y + \frac{i \times i}{i \times -i} \cosh x \sin y.$

$u+iv = \sinh x \cos y + \frac{i}{1} \cosh x \sin y.$

$u+iv = \sinh x \cos y + i \cosh x \sin y.$

$u = \sinh x \cos y \longleftarrow 1 \qquad v = \cosh x \sin y \longleftarrow 2$

From 1 & 2 eliminate x

$\cosh^2 x - \sinh^2 x = 1$

$\sinh x = \frac{u}{\cos y}$

$\cosh x = \frac{v}{\sin y}.$

$\cosh^2 x - \sinh^2 x = 1$

$\frac{v^2}{\sin^2 y} - \frac{u^2}{\cos^2 y} = 1$

which represents a hyperbola

$\sin^2 x + \cos^2 x = 1$

$$\frac{u^2}{\sinh^2 x} + \frac{v^2}{\cosh^2 x} = 1$$

which represents an ellipse.

(v) Discuss the transformation $w = \cosh z$

$u + iv = \cosh(x + iy)$

$u + iv = \cos i(x + iy)$ $\left[\cos(A-B) = \cos A \cos B + \sin A \sin B\right]$

$u + iv = \cos(ix - y)$

$u + iv = \cos ix \cos y + \sin ix \sin y$

$u + iv = \cosh x \cos y + i \sinh x \sin y$

$u = \cosh x \cos y$ — ① $v = \sinh x \sin y$ — ②

From ① & ② eliminate x

$\cosh x = \frac{u}{\cos y}$

$\sinh x = \frac{v}{\sin y}$

$\cosh^2 x - \sinh^2 x = 1$

$$\frac{u^2}{\cos^2 y} - \frac{v^2}{\sin^2 y} = 1$$

which represents a hyperbola.

From ① & ② eliminate y.

$\cos y = \frac{u}{\cosh x}$

$\sin y = \frac{v}{\sinh x}$

$\sin^2 y + \cos^2 y = 1$

$$\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sinh^2 x} = 1$$

which represents an ellipse

$\sin^2 x + \cos^2 x = 1$

$$\frac{u^2}{\sinh^2 x} + \frac{v^2}{\cosh^2 x} = 1$$

which represents an ellipse

10) Discuss the transformation $w = \cosh z$

$u + iv = \cosh(x + iy)$

$u + iv = \cos i(x + iy)$

$u + iv = \cos(ix - y)$ $\left[\begin{array}{l} \cos(A-B) = \cos A \cos B \\ + \sin A \sin B \end{array} \right]$

$u + iv = \cos ix \cos y + \sin ix \sin y$

$u + iv = \cosh x \cos y + i \sinh x \sin y$

$u = \cosh x \cos y$ — ① $v = \sinh x \sin y$ — ②

From ① & ② eliminate x.

$$\cosh x = \frac{u}{\cos y}$$

$$\sinh x = \frac{v}{\sin y}$$

$\cosh^2 x - \sinh^2 x = 1$

$$\frac{u^2}{\cos^2 y} - \frac{v^2}{\sin^2 y} = 1$$

which represents a hyperbola

From ① & ② eliminate y.

$$\cos y = \frac{u}{\cosh x}$$

$$\sin y = \frac{v}{\sinh x}$$

$\sin^2 y + \cos^2 y = 1$

$$\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sinh^2 x} = 1$$

which represents an ellipse

ii) Transformation $w = z^2$

Replace $w = u + iv$

$z = x + iy$

$u + iv = (x+iy)^2$

$u + iv = x^2 + (iy)^2 + 2ixy$

$u + iv = x^2 - y^2 + i\,2xy$

$u = x^2 - y^2$ —— (1) $\qquad v = 2xy$ —— (2)

Case(i):

Take $x = 0$.

Replace $x = 0$ in (1) & (2), we get

$u = -y^2$

$v = 0$

The Imaginary axis $x = 0$ in the z plane is mapped negative u-axis in the w plane.

Case(ii):

Take $y = 0$

From (1) & (2)

$u = x^2$

$v = 0$

The real axis $y = 0$ in the z plane is mapped into positive u-axis in the w plane.

Case(iii):

Take $x=c$

Sub $x=c$ in ① & ② we get

$u = c^2 - y^2$ — ③

$v = 2cy$

$y = \frac{v}{2c}$ — ④

Sub ④ in ③

$u = c^2 - \left(\frac{v}{2c}\right)^2$

$u = c^2 - \frac{v^2}{4c^2}$

$u = \frac{4c^4 - v^2}{4c^2}$

$4uc^2 = 4c^4 - v^2$

$v^2 = -4uc^2 + 4c^4$

$v^2 = 4c^2(c^2 - u)$

$v^2 = -4c^2(u - c^2)$

This represents a parabola of centre $(c^2, 0)$

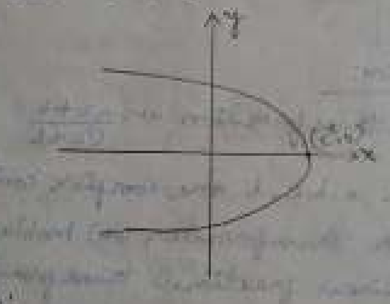
$y^2 = 4ax$

$(y-0)^2 = 4a(x-0)$ centre $(0,0)$

||ly

$v^2 = -4c^2(u-c^2)$

$(v-0)^2 = -4c^2(u-c^2)$ centre $(c^2,0)$



Case(iv):

Take $y=d$.

Sub $y=d$ in ① & ②

$u = x^2 - d^2$ — ⑤

$v = 2xd$

$d = \frac{v}{2x}$ — ⑥

$x = \frac{v}{2d}$ — ⑥

Sub ⑤ in ⑥

$$u = \left(\frac{v}{2d}\right)^2 - d^2$$

$$u = \frac{v^2}{4d^2} - d^2$$

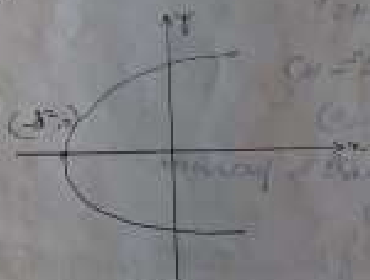
$$u = \frac{v^2 - 4d^4}{4d^2}$$

$$4ud^2 = v^2 - 4d^4$$

$$v^2 = 4ud^2 + 4d^4$$

$$v^2 = 4d^2(u + d^2)$$

It represents a parabola of centre $(-d^2, 0)$



2m

③ Bilinear transformation:

The transformation $w = \frac{az+b}{cz+d}$

where $ad - bc \neq 0$ and a, b, c, d are complex constants is called the Bilinear transformation (or) Mobius transformation (or) Linear fractional transformation

Fixed points (or) invariant points:

The invariant points of the transformation are such that the image of z itself [(i.e.) $w=z$]

cross ratio property of a bilinear transformation

If 3 w points and 3 z points are given the formula is

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

you should bring in the form of $w=\frac{az+b}{cz+d}$

Problems:

1) Find the invariant points of $w=\frac{1+z}{1-z}$

Replace w by z.

$z=\frac{1+z}{1-z}$

$z(1-z)=1+z$

$z-z^2=1+z$

$\cancel{z}-z^2-1-\cancel{z}=0$

$z^2=-1$

$z=\pm\sqrt{-1}$

$\boxed{z=\pm i}$

2) Find the invariant points $w=\frac{6z-9}{z}$

Replace w by z

$z=\frac{6z-9}{z}$

$z^2-6z+9=0$

$(z-3)(z-3)=0$

$\boxed{z=3}$ $\boxed{z=3}$

$z=3,3$ are the invariant points

3) Find the invariant points $w = \frac{2z-5}{z+4}$

$z = \frac{2z-5}{z+4}$

$z^2 + 4z = 2z - 5$

$z^2 + 4z - 2z + 5 = 0.$

$z^2 + 2z + 5 = 0.$

$a = 1, \; b = 2, \; c = 5$

$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-2 \pm \sqrt{4 - 4(1)(5)}}{2(1)}$

$= \frac{-2 \pm \sqrt{4 - 20}}{2}$

$= \frac{-2 \pm \sqrt{-16}}{2}$

$= \frac{-2 \pm 4i}{2}$

$\boxed{z = -1 \pm 2i}$

4) Find the invariant points $w = \frac{z-1}{z+1}$

$z = \frac{z-1}{z+1}$

$z^2 + z = z - 1$

$z^2 + \cancel{z} - \cancel{z} + 1 = 0$

$z^2 = -1$

$\boxed{z = \pm i}$

Type 1:

Find the bilinear transformation $z = (1, i, -1)$, $w = (2, i, -2)$

WKT

The formula for cross ratio is

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$z_1 = 1$ $\quad$ $w_1 = 2$

$z_2 = i$ $\quad$ $w_2 = i$

$z_3 = -1$ $\quad$ $w_3 = -2$

$$\frac{(w-2)(i+2)}{(2-i)(-2-w)} = \frac{(z-1)(i+1)}{(1-i)(-1-z)}$$

$$\frac{(w-2)(i+2)}{-(2+w)(2-i)} = \frac{(z-1)(i+1)}{(1-i)(-(1+z))}$$

$$\frac{(w-2)(i+2)(2+i)}{(2+w)(2-i)(2+i)} = \frac{(z-1)(i+1)(1+i)}{(1-i)(1+z)(1+i)}$$

$$\frac{(w-2)(i+2)^2}{(2+w)(2^2-i^2)} = \frac{(z-1)(1+i)^2}{(1+z)(1^2-i^2)}$$

$$\frac{(w-2)(i^2+4+4i)}{(w+2)(4+1)} = \frac{(z-1)(1^2+i^2+2i)}{(z+1)(1+1)}$$

$$\frac{(w-2)(-1+4+4i)}{(w+2)\,5} = \frac{(z-1)}{(z+1)} \times \frac{\cancel{2}i}{\cancel{2}}$$

$$\frac{(w-2)(4i+3)}{5(w+2)} = \frac{i(z-1)}{(z+1)}$$

$(w-2)(4i+3)(z+1) = 5i(z-1)(w+2)$

$(w-2)(4i+4iz+3z+3) = (z-1)(10i+5iw)$

$w4i + 4izw + 3zw + 3w - 8i - 8iz - 6z - 6$

$= 10iz + 5iwz - 10i - 5iw$

$4iw + 4iwz + 3w + 3wz - 5iwz + 5iw$

$= 18zi - 10i + 8i + 8iz + 6 + 6z$

$9iw - iwz + 3w + 3wz = 18zi - 2i + 6 + 6z$

$w(9i - iz + 3 + 3z) = 18zi - 2i + 6 + 6z$

$$w = \frac{(18zi - 2i + 6 + 6z)}{(9i - iz + 3 + 3z)}$$

$$w = \frac{(18zi + 6z + 6 - 2i)}{3z - iz + 9i + 3}$$

$$w = \frac{z(18i + 6) + (6 - 2i)}{z(3 - i) + (3 + 9i)}$$

$$w = \frac{za + b}{zc + d}$$

which is the required equation

2) Find the bilinear transformation $z = (1, i, -1)$ onto $w = (i, 0, -i)$.

$z_1 = 1$ $\quad w_1 = i$

$z_2 = i$ $\quad w_2 = 0$

$z_3 = -1$ $\quad w_3 = -i$

The formula for cross ratio is

$$\frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\frac{(w - i)(0 + i)}{(i - 0)(-i - w)} = \frac{(z - 1)(i + 1)}{(1 - i)(-1 - z)}$$

$$\frac{(w-i)\cancel{2}}{\cancel{2}(-i-w)} = \frac{(z-1)(1+i)}{(-z-1)(1-i)}$$

$$\frac{(w-i)}{-(w+i)} = \frac{(z-1)(1+i)}{-(z+1)(1-i)}$$

$$\frac{(w-i)}{(w+i)} = \frac{(z-1)(1+i)(1+i)}{(z+1)(1-i)(1+i)}$$

$$\frac{(w-i)}{(w+i)} = \frac{(z-1)}{(z+1)}\frac{(1+i)^2}{(1+1)}$$

$$\frac{(w-i)}{(w+i)} = \frac{(z-1)}{(z+1)} \times \frac{1+i^2+2i}{2}$$

$$\frac{(w-i)}{(w+i)} = \frac{(z-1)}{(z+1)} \times \frac{\cancel{1}\cancel{-1}+\cancel{2}i}{\cancel{2}}$$

$$\frac{(w-i)}{(w+i)} = i\,\frac{(z-1)}{(z+1)}$$

$$(w-i)(z+1) = i(z-1)(w+i)$$

$$wz+w-iz-i = i(wz+iz-w-i)$$

$$wz+w-iz-i = iwz+i^2z-iw-i^2$$

$$wz+w-iz+i = iwz-z-iw+1$$

$$wz+w-iwz+iw = -z+1+iz+i$$

$$w(z+1-iz+i) = -z+iz+1+i$$

$$w = \frac{(-z+iz+1+i)}{(z+1-iz+i)}$$

$$w = \frac{z(i-1)+(1+i)}{z(1-i)+(1+i)} \Rightarrow \frac{za+b}{zc+d}$$

which is the required equation

Type 2:

Find the bilinear transformation which maps $z = (0, 1, \infty)$ onto the w plane $(i, 1, -i)$

(a)

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$w_1 = i,$ $\quad z_1 = 0$

$w_2 = 1$ $\quad z_2 = 1$

$w_3 = -i$ $\quad z_3 = \infty$

$$\frac{(w-i)(1+i)}{(i-1)(-i-w)} = \frac{(z-0)(1-z_3)}{(0-1)(z_3-z)}$$

$$\frac{(w-i)(1+i)}{(i-1)(-i-w)} = \frac{z \cdot \cancel{z_3}\left(\frac{1}{z_3}-1\right)}{(-1)\cancel{z_3}\left(1-\frac{z}{z_3}\right)}$$

Take z_3 common in order to substitute $\frac{1}{\infty} = 0$

$$\frac{(w-i)(1+i)}{(i-1)(-i-w)} = \frac{z \cdot \left(\frac{1}{\infty}-1\right)}{-\left(1-\frac{z}{\infty}\right)}$$

$$= \frac{z(0-1)}{-(1-0)}$$

$$\frac{(w-i)(1+i)}{(-i-w)(i-1)} = \frac{-z}{-1}$$

$$\frac{(w-i)(1+i)(1+i)}{(-i-w)(i-1)(i+1)} = z$$

$$\frac{(\omega-i)\,(i+1)^2}{(-i-\omega)\,(1+1)} = z.$$

$$\frac{(\omega-i)}{(-i-\omega)}\,\frac{i^2+1+2i}{(2)} = z$$

$$\frac{(\omega-i)}{(-i-\omega)} \times \frac{\cancel{-1}+\cancel{1}+\cancel{2}i}{\cancel{2}} = z$$

$$\frac{i\,(\omega-i)}{(-i-\omega)} = z$$

$$z = \frac{(i\omega - i^2)}{(-i-\omega)}$$

$$z = \frac{(i\omega + 1)}{(-i-\omega)}$$

$$z(-i-\omega) = (i\omega + 1)$$

$$-iz - z\omega = i\omega + 1$$

$$\cancel{z(-i-\omega)}$$

$$-z\omega - i\omega = 1 + iz$$

$$\omega(-i-z) = (1+iz)$$

$$\omega = \frac{(1+iz)}{(-i-z)}$$

$$\omega = \frac{(zi)+(1)}{(-z)-(i)}$$

1) Find the bilinear transformation $(0, 1, \infty)$ onto ~~$(0, 1, \infty)$~~ $(1, i, -1)$

$z_1 = 0$ $\quad w_1 = 1$

$z_2 = 1$ $\quad w_2 = i$

$z_3 = \infty$ $\quad w_3 = -1$

$$\frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\frac{(w-1)(i+1)}{(1-i)(-1-w)} = \frac{(z-0)(1-z_3)}{(0-1)(z_3-z)}$$

$$\frac{(w-1)(1+i)}{(-1-w)(1-i)} = \frac{\cancel{z_3}\,(z)\left(\frac{1}{z_3}-1\right)}{(-1)\cancel{z_3}\left(1-\frac{z}{z_3}\right)}$$

$$\frac{(w-1)(1+i)}{(-1-w)(1-i)} = \frac{z\left(\frac{1}{\infty}-1\right)}{-\left(1-\frac{z}{\infty}\right)}$$

$$\frac{(w-1)(1+i)}{(-1-w)(1-i)} = \frac{z(0-1)}{-(1-0)}$$

$$\frac{(w-1)(1+i)}{(-1-w)(1-i)} = z$$

$$\frac{(w-1)}{(-1-w)} \times \frac{(1+i)(1+i)}{(1-i)(1+i)} = z$$

$$\frac{(w-1)}{(-1-w)} \times \frac{(1+i)^2}{(1+1)} = z$$

$$\frac{(w-1)}{(-1-w)} \times \frac{(\cancel{1}-\cancel{1}+2i)}{\cancel{2}} = z$$

$$\frac{i(w-1)}{(-1-w)} = z$$

$$(iw - i) = -z - zw.$$

$$iw + zw = -z + i$$

$$w(z+i) = (-z+i)$$

$$w = \frac{(-z+i)}{(z+i)}$$

14.07.15
Tuesday
11am

Type 3:

10 times

(2) Find the bilinear transformation which transforms $z = (\infty, i, 0)$ on to $(0, i, \infty)$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-0)(i-w_3)}{(0-i)(w_3-w)} = \frac{(z-z_1)(i-0)}{(z_1-i)(0-z)}$$

$$\frac{(w)\,\cancel{w_3}\left(\frac{i}{w_3}-1\right)}{(-i)\cancel{w_3}\left(1-\frac{w}{w_3}\right)} = \frac{\cancel{z_1}\left(\frac{z}{z_1}-1\right)(i)}{\cancel{z_1}\left(1-\frac{i}{z_1}\right)(-z)}$$

$$\frac{w(0-1)}{-i(1-0)} = \frac{(0-1)\,i}{(1-0)(-z)}$$

$$\frac{w(-1)}{-i(1)} = \frac{(-1)\,i}{(-z)}$$

$$w = \frac{i^2}{z}$$

1) Find the bilinear transformation which transforms

$z = (\infty, i, 0)$ onto $(0, -i, \infty)$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-0)(i-w_3)}{(0+i)(w_3-w)} = \frac{(z-z_1)(i-0)}{(z_1-i)(0-z)}$$

$$\frac{(w)\,\cancel{w_3}\left(-\frac{i}{w_3}-1\right)}{(i)\,\cancel{w_3}\left(1-\frac{w}{w_3}\right)} = \frac{\cancel{z_1}\left(\frac{z}{z_1}-1\right)(i)}{\cancel{z_1}\left(1-\frac{i}{z_1}\right)(-z)}$$

$$\frac{w(-1)}{i(1)} = \frac{(-1)(i)}{(1)(-z)}$$

$$-w = \frac{i^2}{z}$$

$$\boxed{w = \frac{1}{z}}$$

2) <u>Taylor Series:</u>

If $f(z)$ is analytic inside a circle (C) with centre at z_0 and radius r, then at each point z inside C is given by $f(z)$

$$f(z) = f(z_0) + \frac{(z-z_0)}{1!}f'(z_0) + \frac{(z-z_0)^2}{2!}f''(z_0) + \frac{(z-z_0)^3}{3!}f'''(z_0) + \cdots \longrightarrow (1)$$

In eqn (1), replace $z_0 = 0$ we get.

$$f(z) = f(0) + \frac{z}{1!} f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \cdots$$

This eqn is called Maclaurin's equation.

Laurent's series:

If C_1 and C_2 are two concentric circles with centre a and radius r_1 and r_2 where $(r_2 < r_1)$ and if $f(z)$ is analytic on C_1 and C_2 and throughout the annular region ~~throughout~~ (R) between them at each point z in R is given by

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=0}^{\infty} \frac{b_n}{(z-a)^n}$$

Problems Partial fractions:

(i) $\frac{p(x)}{(x+a)(x-b)} = \frac{A}{(x+a)} + \frac{B}{(x-b)}$

(ii) $\frac{\cancel{p(x)}}{(x+a)^2(x-b)} = \frac{Ax+B}{(x+a)^2} + \cancel{C}$

(ii) $\frac{p(x)}{(x+a)^2(x-b)} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x-b)}$

(iii) $\frac{p(x)}{(x^2+ax+b)(x-b)} = \frac{Ax+B}{(x^2+ax+b)} + \frac{C}{(x-b)}$

Note (1):

1. If the powers of the numerator and denominator are same then you should first divide and bring in the form of $\frac{p(x)}{\text{divisor}} = \text{Quotient} + \frac{\text{Remainder}}{\text{divisor}}$

Note:
2 the evaluation of coefficients in Taylor's and Laurent's series by means of complex integration is tedious job at the time we will use the binomial series formula $(1+z)^{-1} = 1 - z + z^2 - z^3 + z^4 + \cdots \quad |z|<1$

$$(1-z)^{-1} = 1 + z + z^2 + z^3 + z^4 + \cdots \quad |z|<1$$

Problems

1) Obtain the Taylor's series for the function

$f(z) = \frac{1}{(z-1)(z-2)}$ in the region $|z|<1$

$$f(z) = \frac{1}{(z-1)(z-2)}$$

$$= \frac{A}{(z-1)} + \frac{B}{(z-2)} \quad \text{—— (1)}$$

$$\frac{1}{(z-1)(z-2)} = \frac{A(z-2) + B(z-1)}{(z-1)(z-2)}$$

$$1 = A(z-2) + B(z-1) \quad \text{—— (2)}$$

Put $z=1$

$$1 = A(1-2) + B(0)$$

$$\boxed{A = -1}$$

Put $z=2$

$$1 = A(0) + B(2-1)$$

$$\boxed{B = 1}$$

sub A & B in (B)

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{(z-2)} \longrightarrow (B)$$

(1) $|z| < 1$

$$f(z) = \frac{1}{(z-1)(z-2)} = \frac{-1}{1\,(z-1)} + \frac{1}{(z-2)}$$

$$= \frac{+1}{+1(1-z)} + \frac{1}{(z-2)}$$

$$= (1-z)^{-1} + \frac{1}{(z-2)}$$

$$= 1 + z + z^2 + z^3 + \cdots + \frac{1}{(z-2)}$$

(2) obtain the Taylor series for the function

$$f(z) = \frac{z^2-1}{(z+2)(z+3)} \qquad |z| < 2,\ |z| < 3$$

$$f(z) = \frac{z^2-1}{z^2+2z+3z+6}$$

$$= \frac{z^2-1}{z^2+5z+6}$$

z^2+5z+6 (divisor)

$$\begin{array}{r} 1 \quad (Q) \\ \hline z^2 - 1 \\ z^2 + 5z + 6 \\ \hline -5z - 7 \quad (R) \end{array}$$

$$f(z) = 1 + \frac{-5z-7}{z^2+5z+6}$$

$$f(z) = 1 + \frac{-5z-7}{(z+2)(z+3)} \longrightarrow (A)$$

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)} \quad \text{--- } (1)$$

$$\frac{(-5z-7)}{(z+2)(z+3)} = \frac{A(z+3)+B(z+2)}{(z+2)(z+3)}$$

$$-5z-7 = A(z+3)+B(z+2) \quad \text{--- } (2).$$

Sub $z=-2$ in (2)

$$-5(-2)-7 = A(-2+3)+B(-2+2)$$

$$10-7 = A(1)+B(0)$$

$$\boxed{A=3}$$

Sub $z=-3$ in (2)

$$-5(-3)-7 = A(-3+3)+B(-3+2)$$

$$+15-7 = A(0)+B(-1)$$

$$\boxed{B=-8}$$

From (1)

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{3}{(z+2)} - \frac{8}{(z+3)} \quad \text{--- } (3)$$

Sub (3) in (A)

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{(z+3)} \quad \text{--- } (B)$$

(iii) $|z| < 2$

$|z| < 2$ $\qquad$ $|z| < 3$

$\frac{|z|}{2} < \frac{2}{2}$ $\qquad$ $\frac{|z|}{3} < \frac{3}{3}$

$\frac{|z|}{2} < 1$ $\qquad$ $\frac{|z|}{3} < 1$

Take 2 & 3 outside in (3) we get.

$$f(z) = 1 + \frac{3}{2\left(\frac{z}{2}+1\right)} - \frac{8}{3\left(\frac{z}{3}+1\right)}$$

$$= 1 + \frac{3}{2}\left(z/2+1\right)^{-1} - \frac{8}{3}\left(\frac{z}{3}+1\right)^{-1}$$

$$= 1 + \frac{3}{2}\left(1+z/2\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$$

$$= 1 + \frac{3}{2}\left[1-\frac{z}{2}+\frac{z^2}{4}-\frac{z^3}{8}+\frac{z^4}{16}-\cdots\right] + \left[-\frac{8}{3}\right]\left[1-\frac{z}{3}+\frac{z^2}{9}-\frac{z^3}{27}\right]$$

$$f(z) = 1 + \frac{3}{2}\left[1-\frac{z}{2}+\frac{z^2}{4}-\frac{z^3}{8}+\cdots\right] - \frac{8}{3}\left[1-\frac{z}{3}+\frac{z^2}{9}-\frac{z^3}{27}+\cdots\right]$$

3) Obtain the Taylor series for the function $f(z) = \frac{1}{(z+1)(z+3)}$ valid for $|z|<1$ and $|z|<3$

$$f(z) = \frac{1}{(z+1)(z+3)} = \frac{A}{(z+1)} + \frac{B}{(z+3)} \quad \text{--- (1)}$$

$$\frac{1}{(z+1)(z+3)} = \frac{A(z+3)+B(z+1)}{(z+1)(z+3)}$$

$$1 = A(z+3) + B(z+1) \quad \text{--- (2)}$$

put $z=-1$

$$1 = A(-1+3) + B(-1+1)$$

$2A = 1$

$$\boxed{A = 1/2}$$

Put $z = -3$

$$1 = A(-3+3) + B(-3+1)$$

$$1 = -2B$$

$$\boxed{B = -1/2}$$

From (1)

$$f(z) = \frac{1}{2(z+1)} - \frac{1}{2(z+3)}$$

$|z| < 1$ $\qquad$ $|z| < 3$

$|z| < 1$ $\qquad$ $\frac{|z|}{3} < 1$

$$f(z) = \frac{1}{2(1+z)} - \frac{1}{2(z+3)}$$

$$= \frac{1}{2(1+z)} - \frac{1}{2\times 3(\frac{z}{3}+1)}$$

$$= \frac{1}{2(1+z)} - \frac{1}{6(1+z/3)}$$

$$= \frac{1}{2}(1+z)^{-1} - \frac{1}{6}(1+z/3)^{-1}$$

$$f(z) = \frac{1}{2}\left[1 - z + z^2 - z^3 + \dots\right] - \frac{1}{6}\left[1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots\right]$$

05.07.15 Wednesday 11.00

1) Find the Laurent's series for $f(z)=\frac{z^2-4}{(z+1)(z+4)}$

$1<|z|<4$

$$f(z)=\frac{z^2-4}{z^2+z+4z+4}$$

$$=\frac{z^2-4}{z^2+5z+4}$$

$$\begin{array}{r|l} & 1 \\ z^2+5z+4 & z^2-4 \\ & z^2+5z+4 \\ & (-)\ (-)\ (-) \\ \hline & -5z-8 \end{array}$$

$$f(z)=1+\frac{-5z-8}{z^2+5z+4}$$

$$f(z)=1+\frac{-5z-8}{(z+1)(z+4)} \quad \text{——} \ A$$

Apply partial fraction on the second part, we get.

$$\frac{-5z-8}{(z+1)(z+4)}=\frac{A}{(z+1)}+\frac{B}{(z+4)} \quad \text{——} \ 1$$

$$\frac{-5z-8}{(z+1)(z+4)}=\frac{A(z+4)+B(z+1)}{(z+1)(z+4)}$$

$$-5z-8=A(z+4)+B(z+1)$$

Put $z=-1$

$$-5(-1)-8=A(-1+4)+B(-1+1)$$

$$-3=3A$$

$$\boxed{A=-1}$$

put $z=-4$.

$$-5(-4)-8=A(0)+B(-4+1)$$

$$12=-3B$$

$$\boxed{B=-4}$$

$$\frac{-5z-8}{(z+1)(z+4)} = \frac{-1}{(z+1)} + \frac{-4}{(z+4)}$$

$$f(z) = 1 - \frac{1}{(z+1)} - \frac{4}{(z+4)} \quad \text{--- (2)}$$

$1 < |z| < 4$

Case (i): $1 < |z|$ Case (ii) $|z| < 4$

$\frac{1}{|z|} < 1$ $\frac{|z|}{4} < 1$

From (2)

$$f(z) = 1 - \frac{1}{z(1+\frac{1}{z})} - \frac{4}{4(\frac{z}{4}+1)}$$

$$= 1 - \frac{1}{z(1+\frac{1}{z})} - \frac{1}{(1+\frac{z}{4})}$$

$$= 1 - \frac{1}{z}(1+1/z)^{-1} - (1+z/4)^{-1}$$

$$f(z) = 1 - \frac{1}{z}\left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \cdots\right) - \left(1 - \frac{z}{4} + \frac{z^2}{16} - \frac{z^3}{64} + \cdots\right)$$

2) Find the Laurent series for $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ valid for (i) $2 < |z| < 3$ (ii) $|z| > 3$

$$f(z) = \frac{z^2-1}{(z+2)(z+3)}$$

$$f(z) = \frac{z^2-1}{z^2+5z+6}$$

$$z^2+5z+6 \overline{\smash{)}\, z^2-1 \,}\,1$$
$$\frac{z^2+5z+6}{-5z-7}$$

$$f(z) = 1 + \frac{-5z-7}{z^2+5z+6}.$$

$$f(z) = 1 + \frac{-5z-7}{z^2+5z+6} \quad \text{—— (A)}$$

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)}$$

$$\frac{(-5z-7)}{(z+2)(z+3)} = \frac{A(z+3) + B(z+2)}{(z+2)(z+3)}$$

$$-5z-7 = A(z+3)+B(z+2) \quad \text{—— (1)}$$

put $z=-3$

$15-7 = A(0) + B(-3+2)$

$8 = -B$

$\boxed{B=-8}$

put $z=-2$

$10-7 = A(-2+3) + B(0)$

$\boxed{A=3}$

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{3}{(z+2)} - \frac{8}{(z+3)}$$

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{(z+3)} \quad \text{—— (B)}$$

(i) $2<|z|<3$

$2<|z|$ $\qquad$ $|z|<3$

$\frac{2}{|z|}<1$ $\qquad$ $\frac{|z|}{3}<1$

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$$f(z) = 1 + \frac{3}{z\left(1+\frac{2}{z}\right)} - \frac{8}{3\left(1+\frac{z}{3}\right)}$$

$$= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$$

$$f(z) = 1 + \frac{3}{z}\left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \cdots\right) - \frac{8}{3}\left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \cdots\right)$$

(ii) $|z| > 3$

$3 < |z|$

$\frac{3}{|z|} < 1$

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{(z+3)}$$

$$= 1 + \frac{3}{(z+2)} - \frac{8}{z\left(1+\frac{3}{z}\right)}$$

$$= 1 + \frac{3}{(z+2)} - \frac{8}{z}\left(1+\frac{3}{z}\right)^{-1}$$

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{z}\left(1 - \frac{3}{z} + \frac{9}{z^2} - \frac{27}{z^3} + \cdots\right)$$

3) Find the Laurent's series for $f(z) = \frac{1}{z^2-3z+2}$

(i) $|z| > 2$ (ii) $1 < |z| < 2$

$$f(z) = \frac{1}{(z^2-3z+2)}$$

$$\frac{1}{(z^2-3z+2)} = \frac{1}{(z-1)(z-2)}$$

$$\frac{1}{(z^2-3z+2)} = \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

$$\frac{1}{z^2-3z+2} = \frac{A(z-2)+B(z-1)}{(z-2)\cancel{(z-1)}}$$

$$1 = A(z-2) + B(z-1)$$

put $z = 2$

$$1 = A(0) + B(2-1)$$

$$1 = B(1)$$

$$\boxed{B=1}$$

put $z=1$

$$1 = A(1-2) + B(0)$$

$$\boxed{A=-1}$$

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{(z-2)} \quad \text{—— } 1$$

(i) $|z| > 2$

$$2 < |z|$$

$$\frac{2}{|z|} < 1$$

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{z(1-\frac{2}{z})}$$

$$= \frac{-1}{(z-1)} + \frac{1}{z}\left(1-\frac{2}{z}\right)^{-1}$$

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{z}\left(1+\frac{2}{z}+\frac{4}{z^2}+\frac{8}{z^3}+ \cdots\right)$$

(ii) $1 < |z| < 2$

$1 < |z| \qquad |z| < 2$

$\frac{1}{|z|} < 1 \qquad \frac{|z|}{2} < 1.$

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{(z-2)}$$

$$= \frac{-1}{z\left(1-\frac{1}{z}\right)} + \frac{1}{2\left(\frac{z}{2}-1\right)}$$

$$= \frac{-1}{z\left(1-\frac{1}{z}\right)} + \frac{1}{-2\left(1-\frac{z}{2}\right)}$$

$$= \frac{-1}{z}\left(1-\frac{1}{z}\right)^{-1} - \frac{1}{2}\left(1-\frac{z}{2}\right)^{-1}$$

$$= -\frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z}+ \cdots\right) - \frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\cdots\right)$$

17.07.15 Friday

2) Find the Laurent series for $f(z) = \frac{7z-2}{z(z+1)(z-2)}$ valid for (i) $1 < |z+1| < 3$ (ii) $|z+1| > 3$

(5) 12 times 11m

Sol

Put $z+1 = u$

$z = u-1$

$$f(u-1) = \frac{7(u-1)-2}{(u-1+1)(u-1)(u-1-2)}$$

$$f(u-1) = \frac{7u-7-2}{u(u-1)(u-3)}$$

$$f(u-1) = \frac{7u-9}{u(u-1)(u-3)} \longleftarrow (1)$$

$$= \frac{7u-9}{u(u-1)(u-3)} = \frac{A}{u} + \frac{B}{u-1} + \frac{C}{u-3}$$

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$$\frac{(7u-9)}{u(u-1)(u-3)} = \frac{A(u-1)(u-3)+Bu(u-3)+Cu(u-1)}{u(u-1)(u-3)}$$

$$7u-9 = A(u-1)(u-3) + Bu(u-3) + Cu(u-1)$$

Put $u=0$

$$-9 = A(-1)(-3) + B(0) + C(0)$$

$$3A = -9$$

$$\boxed{A = -3}$$

Put $u=3$

$$7(3)-9 = A(0) + B(0) + C(3)(3-1)$$

$$21-9 = 6C$$

$$12 = 6C$$

$$\boxed{C = 2}$$

Put $u=1$

$$7(1)-9 = A(0) + B(1)(1-3) + C(0).$$

$$-2 = -2B$$

$$\boxed{B=1}$$

$$\frac{7u-9}{u(u-1)(u-3)} = \frac{-3}{u} + \frac{1}{(u-1)} + \frac{2}{(u-3)} \quad \text{——} (2)$$

(a) $1 < |z+1| < 3$

$1 < |u| < 3$

$1 < |u| \qquad |u| < 3$

$\frac{1}{|u|} < 1 \qquad \frac{|u|}{3} < 1$

$$f(u-1) = \frac{-3}{u} + \frac{1}{(u-1)} + \frac{2}{(u-3)}$$

$$= \frac{-3}{u} + \frac{1}{u(1-\frac{1}{u})} + \frac{2}{3(\frac{u}{3}-1)}$$

$$= \frac{-3}{u} + \frac{1}{u(1-\frac{1}{u})} - \frac{2}{3(1-u/3)}$$

$$= -\frac{3}{u} + \frac{1}{u}(1-1/u)^{-1} - \frac{2}{3}(1-u/3)^{-1}$$

$$f(u-1) = -\frac{3}{u} + \frac{1}{u}\left(1+\frac{1}{u}+\frac{1}{u^2}+\frac{1}{u^3}+\cdots\right) - \frac{2}{3}\left(1+\frac{u}{3}+\frac{u^2}{9}+\frac{u^3}{27}+\cdots\right.$$

$$f(z) = \frac{-3}{(z+1)} + \frac{1}{(z+1)}\left(1+\frac{1}{(z+1)}+\frac{1}{(z+1)^2}+\frac{1}{(z+1)^3}+\cdots\right) - \frac{2}{3}\left(1+\frac{(z+1)}{3}+\cdots\right.$$

(ii) $|z+1| > 3$

$3 < |z+1|$

$3 < |u|$

$\frac{3}{|u|} < 1$

$$f(u-1) = -\frac{3}{u} + \frac{1}{(u-1)} + \frac{2}{(u-3)}$$

$$= -\frac{3}{u} + \frac{1}{(u-1)} + \frac{2}{u(1-\frac{3}{u})}$$

$$= \frac{-3}{u} + \frac{1}{(u-1)} + \frac{2}{u}(1-3/u)^{-1}$$

$$= -\frac{3}{u} + \frac{1}{(u-1)} + \frac{2}{u}\left(1+\frac{3}{u}+\frac{9}{u^2}+\frac{27}{u^3}+\cdots\right)$$

$$f(z) = \frac{-3}{z+1} + \frac{1}{z} + \frac{2}{(z+1)}\left(1+\frac{3}{(z+1)}+\frac{9}{(z+1)^2}+\frac{27}{(z+1)^3}+\cdots\right)$$

3) Find the Laurent series for $f(z) = \frac{z}{(z-1)(z-2)}$

for $1 < |z+2| < 2$.

Let $z+2 = u$

$$\boxed{z = u-2}$$

$$f(u-2) = \frac{(u-2)}{(u-2+1)(u-2-2)}$$

$$f(u-2) = \frac{(u-2)}{(u-3)(u-4)}$$

$$\frac{(u-2)}{(u-3)(u-4)} = \frac{A}{(u-3)} + \frac{B}{(u-4)}$$

$$\frac{(u-2)}{\cancel{(u-3)(u-4)}} = \frac{A(u-4)+B(u-3)}{\cancel{(u-3)(u-4)}}$$

Put $u=4$

$4-2 = A(0) + B(1)$

$$\boxed{B=2}$$

Put $u = z$

$3 - 2 = A(-1) + B(0)$

$\boxed{A = -1}$

$$f(u-2) = \frac{-1}{(u-3)} + \frac{2}{(u-4)}$$

$1 < |z+2| < 2$

$1 < |u| < 2$

$1 < |u|$ $\qquad\qquad$ $|u| < 2$

$\frac{1}{|u|} < 1$ $\qquad\qquad$ $\frac{|u|}{2} < 1$

$$f(u-2) = \frac{-1}{(u-3)} + \frac{2}{(u-4)}$$

The given condition is not valid so we cannot expand in binomial series.

Types of singularities:

There are three types of singularities

(i) Isolated singularitys

(ii) Essential singularity

(iii) Removable singularity

Isolated singularity

A point $z=a$ is said to be an isolated singularity of $f(z)$

(i) if $z=a$ should be a singular point

(ii) if the neighbourhood of $z=a$ should not contain any other singular point.

Removable singularity:

A point $z=a$ is called a removable singularity of $f(z)$

(i) if $z=a$ is a singular point

(ii) $\lim_{z \to a} f(z)$ exists.

Essential singularity:

A point $z=a$ is an essential singularity if

(i) $z=a$ is a singular point

(ii) $z=a$ should not be a removable singularity.

) Find the Laurent series for $f(z) = \frac{4z+1}{z(z^2+z-2)}$

$2 < |z+2| < 3$.

Let $z+2 = u$

$z = u-2$

$$f(u-2) = \frac{4(u-2)+1}{(u-2)((u-2)^2+u-2-2)}$$

$$= 4u-8$$

$$f(z) = \frac{4z+1}{z(z+2)(z-1)}$$

$$f(u-2) = \frac{4(u-2)+1}{(u-2)(u-\cancel{2}+\cancel{2})(u-2-1)}$$

$$f(u-2) = \frac{4u-8+1}{(u-2)\,u\,(u-3)}$$

$$f(u-2) = \frac{4u-7}{u(u-2)(u-3)}$$

$$\frac{4u-7}{u(u-2)(u-3)} = \frac{A}{u} + \frac{B}{(u-2)} + \frac{C}{(u-3)}$$

$$\frac{4u-7}{\cancel{u(u-2)(u-3)}} = \frac{A(u-2)(u-3)+Bu(u-3)+Cu(u-2)}{\cancel{u(u-2)(u-3)}}$$

$$4u-7 = A(u-2)(u-3)+Bu(u-3)+Cu(u-2)$$

put $u=0$

$$-7 = A(-2)(-3) + B(0) + C(0)$$

$$\boxed{A = -7/6}$$

put $u=3$

$$12-7 = A(0) + B(0) + C(3)(1)$$

$$5 = 3C$$

$$\boxed{C = 5/3}$$

Put $u=2$

$8-7 = A(0) + B(2)(-1) + C(0).$

$1 = -2B$

$\boxed{B = -1/2}$

$$\frac{4u-7}{u(u-2)(u-3)} = \frac{-7}{6(u)} - \frac{1}{2(u-2)} + \frac{5}{3(u-3)}$$

$$f(u-2) = \frac{-7}{6u} - \frac{1}{2(u-2)} + \frac{5}{3(u-3)}$$

$2<|u|<3.$